

Name: _____**Rules:**

You have 90 minutes to complete this midterm.

Partial credit will be awarded, but you must show your work.

Calculators are not allowed.

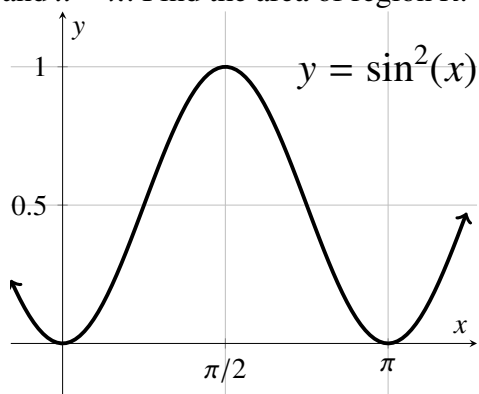
Place a box around your FINAL ANSWER to each question where appropriate.

Turn off anything that might go beep during the exam.

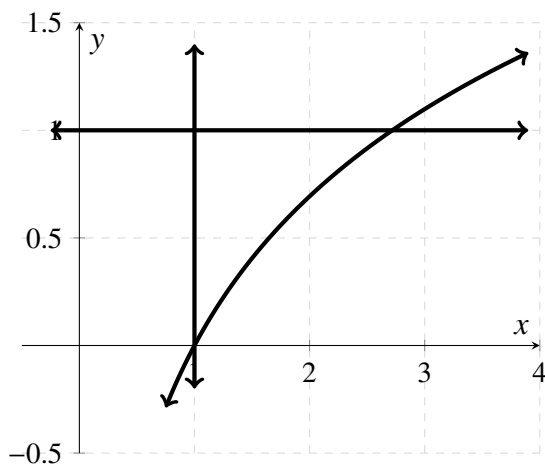
Good luck!

Problem	Possible	Score
1	12	
2	16	
3	8	
4	11	
5	10	
6	11	
7	32	
Extra Credit	5	
Total	100	

1. (12 points) The region R is bounded by $y = \sin^2(x)$ (graphed below) and the x -axis between $x = 0$ and $x = \pi$. Find the area of region R .



2. (16 points) The region R , sketched below, is bounded by $y = \ln(x)$, $x = 1$ and $y = 1$. **Set up but do not evaluate** a definite integral to calculate each of the volumes described below.



- (a) Use the **slicing (disks/washers)** method to find the volume generated when R is rotated about the **x -axis**. (Set up only.)
- (b) Use the **shell** method to find the volume generated when R is rotated about the **x -axis**. (Set up only.)
- (c) Use which ever method you prefer to find the volume generated when R is rotated about the **y -axis**. (Set up only.) **State the method you are using.**

3. (8 points) Set up but do not evaluate integrals to find the surface area of the volume generated when the curve $y = x^2$ from $x = 0$ to $x = 3$ is rotated about

(a) the x -axis.

(b) the y -axis.

4. (11 points) A 3-meter long whip antenna has linear density $\rho(x) = 5 \ln(x)$ grams per meter for x -values in the interval $[1, 4]$. Determine the mass of the antenna. Include units.

5. (10 points)

- (a) If 30 Nm of work are required to compress a spring from 5 meters (neutral length) to 4 meters, determine the spring constant k in Hooke's Law.

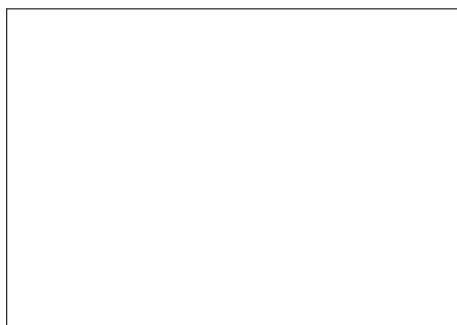
- (b) For the spring in part a , how far must the spring be compressed from its natural position if the work required to do so is 120 Nm?

6. (11 points) Let R be the region bounded by $y = x^2$ and $y = x$.

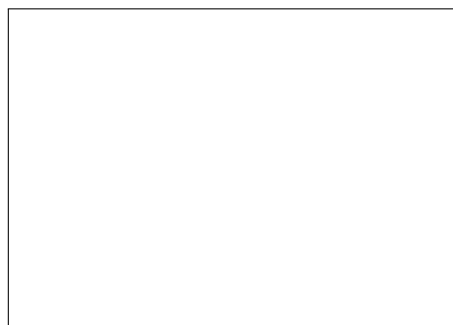
(a) (3 points) Sketch the region R .

(b) (6 points) Set up the definite integrals to compute the center of mass (centroid) $(\bar{x}, \bar{y})$ of R . You do not have to evaluate the integrals.

$\bar{x} =$



$\bar{y} =$



(c) (2 points) Suppose someone calculates the center of mass of R to be $(\bar{x}, \bar{y}) = (0.8, 0.8)$. Is this plausible? You can and should answer this **without actually evaluating the integrals in part (b)**.

7. (8 points each) Evaluate the indefinite integrals.

(a) $\int \cos^3(x) \sin^3(x) \, dx$

(b) $\int x e^{x/2} \, dx$

(c) $\int \frac{x+4}{x^2+x-6} dx$

(d) $\int \frac{\sqrt{x^2-25}}{x} dx$

Extra Credit (5 points) Evaluate the integral $\int x \sin^2(x) \cos(x) dx$

Half-Angle/Double Angle	Sum and Difference
$\sin^2(x) = \frac{1}{2}(1 - \cos(2x))$	$\sin(ax) \cos(bx) = \frac{1}{2}(\sin((a - b)x) + \sin((a + b)x))$
$\cos^2(x) = \frac{1}{2}(1 + \cos(2x))$	$\sin(ax) \sin(bx) = \frac{1}{2}(\cos((a - b)x) - \cos((a + b)x))$
$\sin(2x) = 2 \sin(x) \cos(x)$	$\cos(ax) \cos(bx) = \frac{1}{2}(\cos((a - b)x) + \cos((a + b)x))$

$$L = \int_a^b \sqrt{1 + (f'(x))^2} dx \qquad S = \int_a^b 2\pi f(x) \sqrt{1 + (f'(x))^2} dx$$